

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘A’

Course Code & Title: EC8393 & Fundamentals of Data Structures in C

Practice's followed: Collaborative Learning

Type of Activity: Student Teams-Achievement Divisions (STAD)

Topic: Tree and Graph Traversal

Date: 19-09-19

Objectives:

O1: To familiarize the students to learn and explore tree and graph data structures.

Justification for choosing the particular topic:

Traversing the tree and graph is more important concept in non-linear data structures. To make the students familiar with pre-order, in-order, post-order, in-degree, out-degree Student Teams-Achievement Divisions (STAD) activity was conducted. This activity encouraged the students to share their knowledge with others. So, the students can clarify their doubts and analyze the mistakes.

STAD:

In Student Teams-Achievement Divisions (STAD) in which students are assigned to five to six member learning teams that are mixed in performance level. The STAD method is most appropriate for teaching well-defined objectives with single right answers, such as applications, map skills and concepts.

Components of STAD:

- Class presentation – Materials introduced first in the classroom presentation. The concepts such as Binary Search Trees, BFS, DFS, pre-order, in-order, post-order, in-degree, out-degree were taught.
- Teams - The students are assigned to four or five member teams that are mixed in heterogeneous teams. Students were divided based on their performance in subject. Each team has the combination of fast learners, average learners and slow learners.
- Quizzes – Each team has given set of quiz question topic under tree and graph traversal.
- Team Work – The students worked with team members to find the answers.

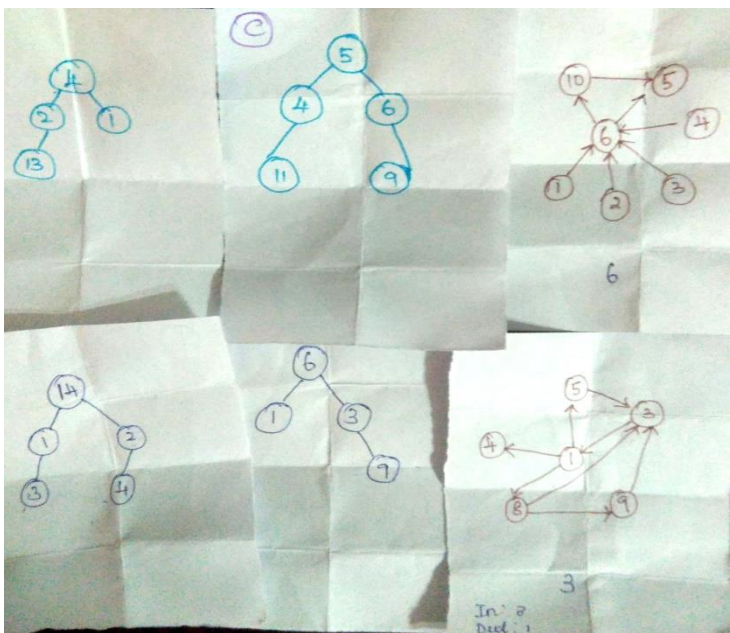
Pre-Plan:

- The activity information is given to the students three days before.
- Advise them to practice/ refresh the tree traversal and graph traversal.
- Discussion about the importance of team study

Implementation:

- For each team, 2 trees and 1 graph were given. Students from each team was insisted to write in-order, pre-order for the given tree and also asked to write in-degree and out-degree for all the vertices in the given graph.
- The concept of the activity was actually to answer the questions which were displayed on the screen. Each team should answer for 3 questions totally 15 questions were answered by them. Based on the traversal find by them, we have posted the quiz questions to them.
- The questions were displayed after finding traversal of tree and graph. If the team 1, answers for in-order traversal starting is 4 then we displayed 4th quiz question.

Likewise all the teams were answered for the displayed question. Finally answers were discussed.



$A \rightarrow$	<u>13</u>	2	4	1	
$B \rightarrow$	<u>15</u>	6	7	9	
$C \rightarrow$	<u>11</u>	4	5	6	9
$D \rightarrow$	<u>9</u>	8	1	10	
$E \rightarrow$	<u>12</u>	3	7	4	9
<u>Includes</u>					
$1 \rightarrow$	3	1	4	2	<u>14</u>
$2 \rightarrow$	16	6	4	15	11 <u>8</u>
$3 \rightarrow$	3	2	4	5	<u>7</u>
$4 \rightarrow$	12	11	4	<u>10</u>	
$5 \rightarrow$	1	9	3	<u>6</u>	
<u>Postorder</u>					
$a \rightarrow$	$I = 2, O = 3$				
$b \rightarrow$	$I = 4, O = 2$				
$e \rightarrow$	$I = 1, O = 4$				
$d \rightarrow$	$I = 3, O = 1$				
$c \rightarrow$	$I = 5, O = 2$				
<u>Indegree</u>					

Reflection critique

Challenges:

- Tough to create different tree and graph representation for each group.
- Most of the students were able to share their learning, some of the students found struggle to traverse the tree.

Post Implementation:

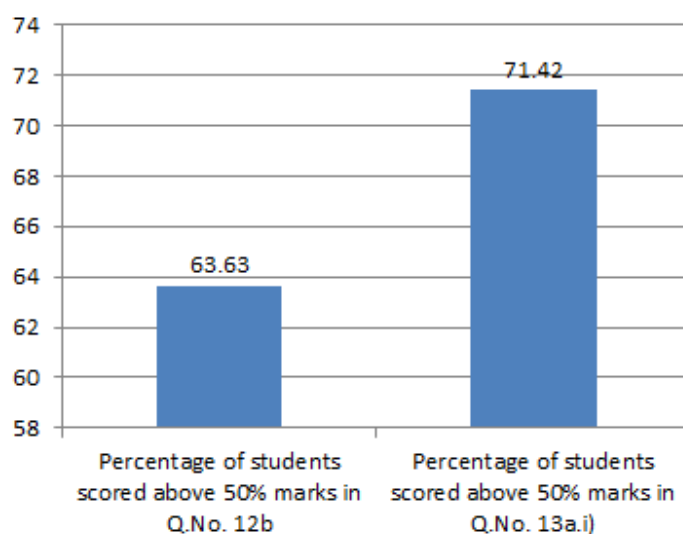
- Identified the different level of learners and analyze the performance.
- Fast learners in the group helped slow learning peer for finding answers.

Significance of Results:

Internal Assessment Test III was conducted on 12.10.2019. The impact of the innovative practice is analyzed in the table below.

Total No. of Students present for IAT III	36
No. of students attended Q.No. 12b in IAT III	33
No. of students attended Q.No. 13a.i) in IAT III	14
No. of students scored above 50% marks in Q.No. 12b	21
Percentage of students scored above 50% marks in Q.No. 12b	63.63
No. of students scored above 50% marks in Q.No. 13a.i)	10
Percentage of students scored above 50% marks in Q.No. 13a.i)	71.42

The following diagram illustrates the total number of students who attempted the question and scored above 50% marks.



Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2	2					2	1		1

References

1. <https://www.youtube.com/watch?v=NP85JtkrV3Y>
2. <https://innovativelearning.com>
3. Dr.M.Kaliappan, "Implementation of Service Oriented Principles course" in course "IT6801 Service Oriented Architecture"
4. Monchai Tiantong and Sanit Teemuangsai, Student Team Achievement Divisions (STAD) Technique through the Moodle to Enhance Learning Achievement, International Education Studies; Vol. 6, No. 4; 2013.

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